5.12.5

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Question

Solve for $x \in \mathbb{N}$ using matrices: $\det \begin{pmatrix} x+3 & -2 \\ -3x & 2x \end{pmatrix} = 8$

Solution

$$\det\begin{pmatrix} x+3 & -2 \\ -3x & 2x \end{pmatrix} = (x+3)(2x) - (-2)(-3x) \tag{1}$$

$$= 2x^2 + 6x - 6x \tag{2}$$

$$= 2x^2 \tag{3}$$

$$2x^2 = 8 \tag{4}$$

$$x^2 = 4 \tag{5}$$

$$x = \pm 2 \tag{6}$$

$$x \in \mathbb{N} \implies x = 2 \tag{7}$$

Python code - To solve

```
import numpy as np
from sympy import symbols, Eq, solve

# Define the variable
x = symbols('x')

# Define the matrix
A = np.array([[x + 3, -2],
              [-3*x, 2*x]])

# Compute the determinant using sympy
det = (x + 3)*(2*x) - (-2)*(-3*x)

# Equation: determinant = 8
equation = Eq(det, 8)
```

Python code - To solve

```
# Solve the equation
solutions = solve(equation, x)

# Filter for natural numbers
natural_solutions = [sol for sol in solutions if sol.is_real and
    sol > 0]

print("All solutions:", solutions)
print("Natural number solution(s):", natural_solutions)
```

Python code - Output

```
All solutions: [-2, 2]
Natural number solution(s): [2]
```

C code - Solving

```c
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
#include "/home/dhanush-kumar-a/ee1030-2025/ai25btech11010/matgeo
    /5.12.5/codes/libs/matfun.h"


// Function to solve determinant equation det([[x+3, -2], [-3*x,
    2]]) = 8
double solve_det_equation() {
    double **A = createMat(2, 2);

    // We'll solve the equation 2*x^2 = 8, but let's use matrix
        det
    // Represent matrix as [[x+3, -2], [-3*x, 2]]
    // We'll solve symbolically by using quadratic formula
    double a = 2.0; // coefficient of x^2
    double b = 0.0; // coefficient of x
    double c = -8.0; // constant term
```

C code - Solving

```
    double x = roots[0][0]; // Return positive root (x  N)

    // Free allocated memory
    free(A);
    free(roots);

    return x;
}

// Entry point for Python ctypes
double solve_det_ctypes() {
    return solve_det_equation();
}
```

Python code -Solving using c Funtion

```
import ctypes

lib = ctypes.CDLL('./libsolve_det.so') # Shared library compiled
    from above C code
lib.solve_det_ctypes.restype = ctypes.c_double

x = lib.solve_det_ctypes()
print("Solution x  N:", x)
```

Python code-Output

```
Solution x  N: 2.0
```